

Managing EKS Cluster Using kubectl

To Begin with the Lab

Summary of the Lab

In this lab, an **EC2** instance was launched in the same **VPC** as the **Amazon EKS** cluster so it could manage the cluster safely. The instance was connected through **SSH**, **AWS CLI** was verified, and authentication was configured using an AWS access key and secret key for the lab setup. The **kubectl** binary was then downloaded, made executable, and moved into the system path so it could be used from anywhere. After that, the kubeconfig file was generated with `aws eks update-kubeconfig`, which allowed **kubectl** to communicate with the EKS API server. The connection was verified with commands like `kubectl cluster-info`, `kubectl get ns`, and `kubectl get pods`, and the lab ended with troubleshooting guidance for security group issues if the connection failed.

Understanding kubectl for EKS Management

kubectl is the official command-line tool for Kubernetes, and it talks directly to the Kubernetes API server. It exists because the **AWS Management Console** can manage cluster settings, access, add-ons, networking, and monitoring, but it cannot create pods, apply YAML, view pod logs, or scale applications directly. In this lab, **kubectl** was installed on an **EC2** instance inside the same **VPC** as the EKS cluster, which makes private endpoint access easier. The benefit is that you can manage Kubernetes workloads properly even when the EKS console only shows the cluster-level view.

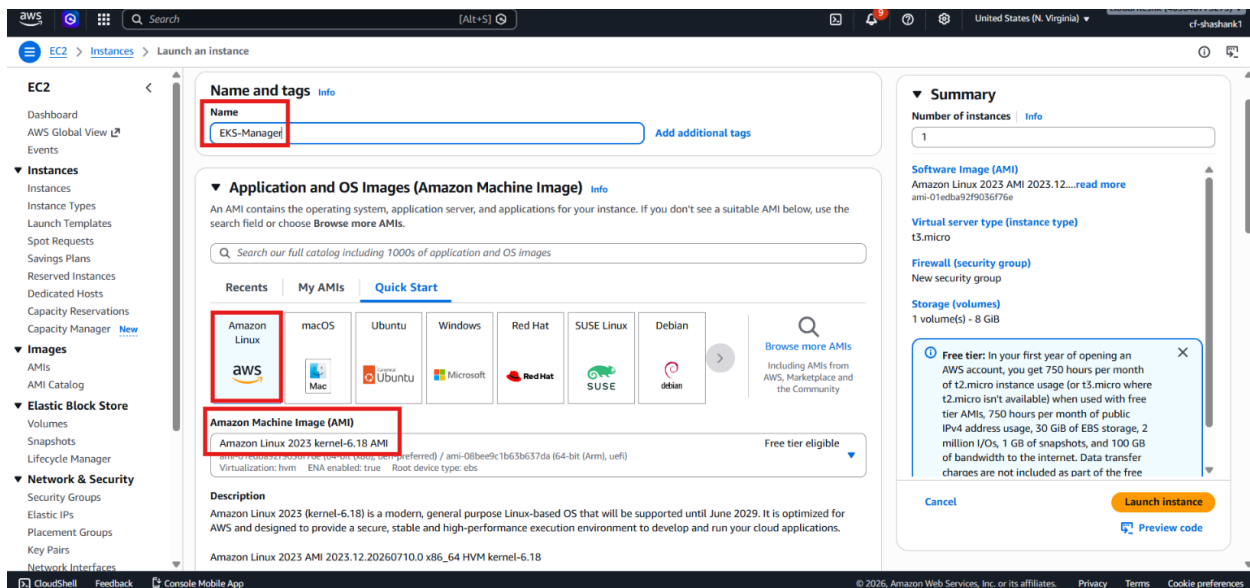
Example:

A developer can use the **EKS console** to check the cluster status and access entries, but if they want to deploy a pod or inspect namespace details, they must use **kubectl**. In this lab, the EC2 instance inside the same VPC was used as the management machine, so it could reach the EKS cluster endpoint and run Kubernetes commands successfully.

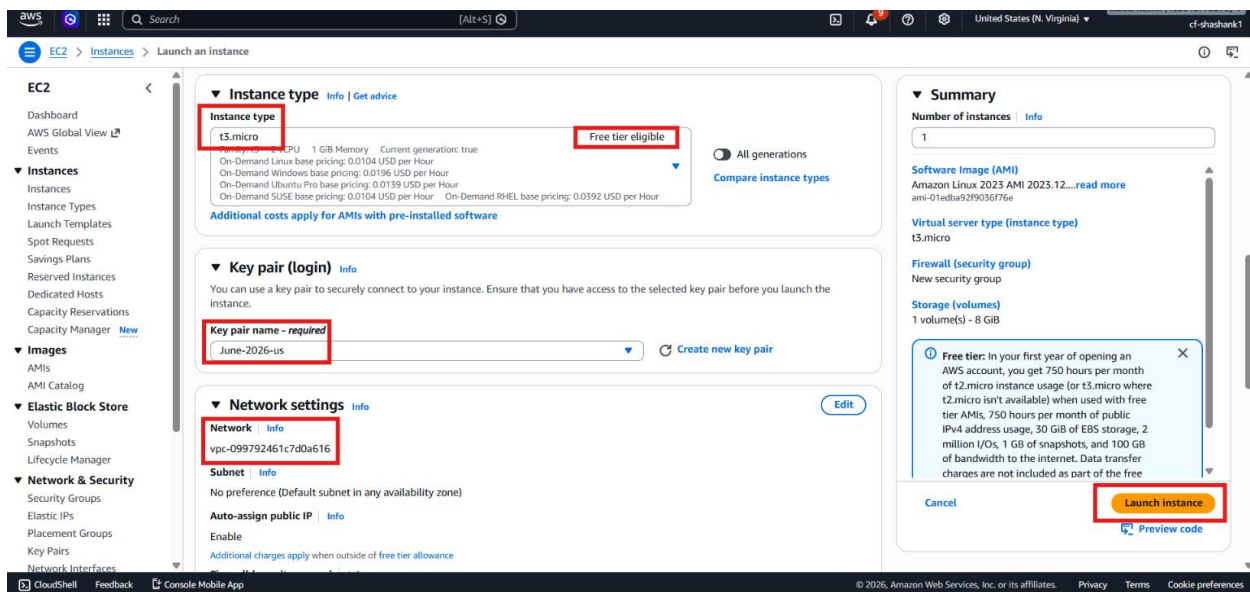
Lab Steps

Step 1 – Launch the management EC2 instance

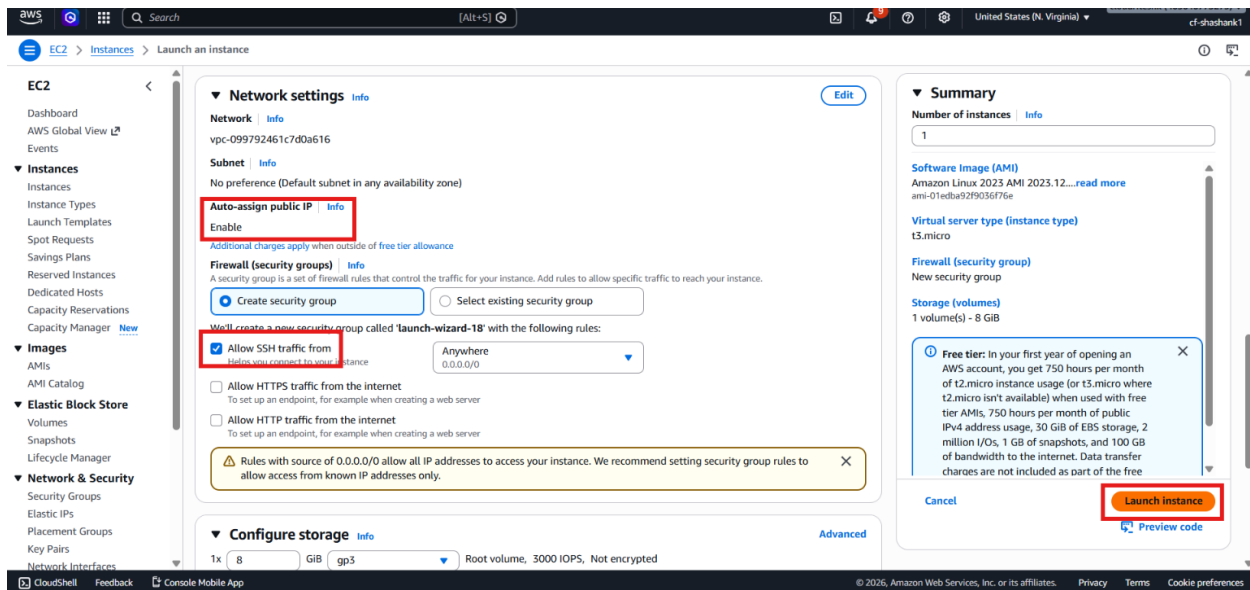
- Sign in to the **AWS Management Console**.
- Open **Amazon EC2**.
- Launch a new instance named **EKS-Manager**.
- Choose **Amazon Linux 2023**.



- Select the t3.micro instance type free tier.
- Use the existing key pair.
- Launch the instance in the same VPC as the EKS cluster.



- Keep SSH inbound open and make sure outbound HTTPS is allowed.



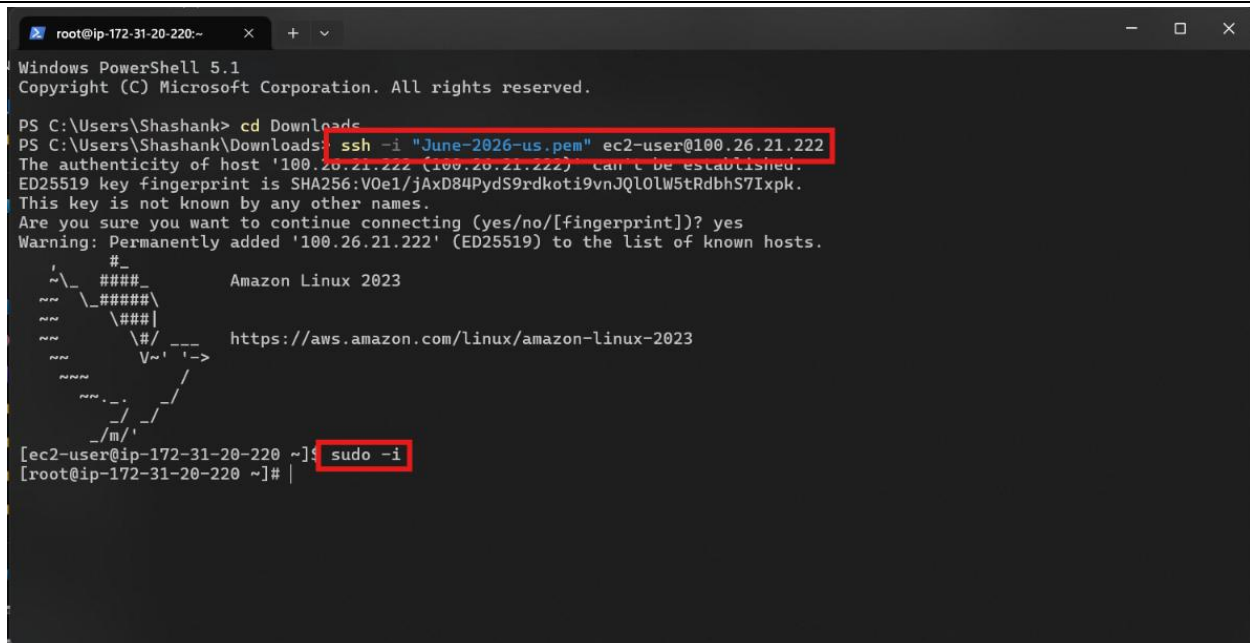
Step 2 – Connect to the EC2 instance

- Open a terminal or command prompt.
- SSH into the EC2 instance using the key pair.
- Log in as the default ec2-user as we did earlier.

```
ssh -i <YOUR_KEY_PAIR_NAME> ec2-user@<PUBLIC_IPV4_ADDRESS>
```

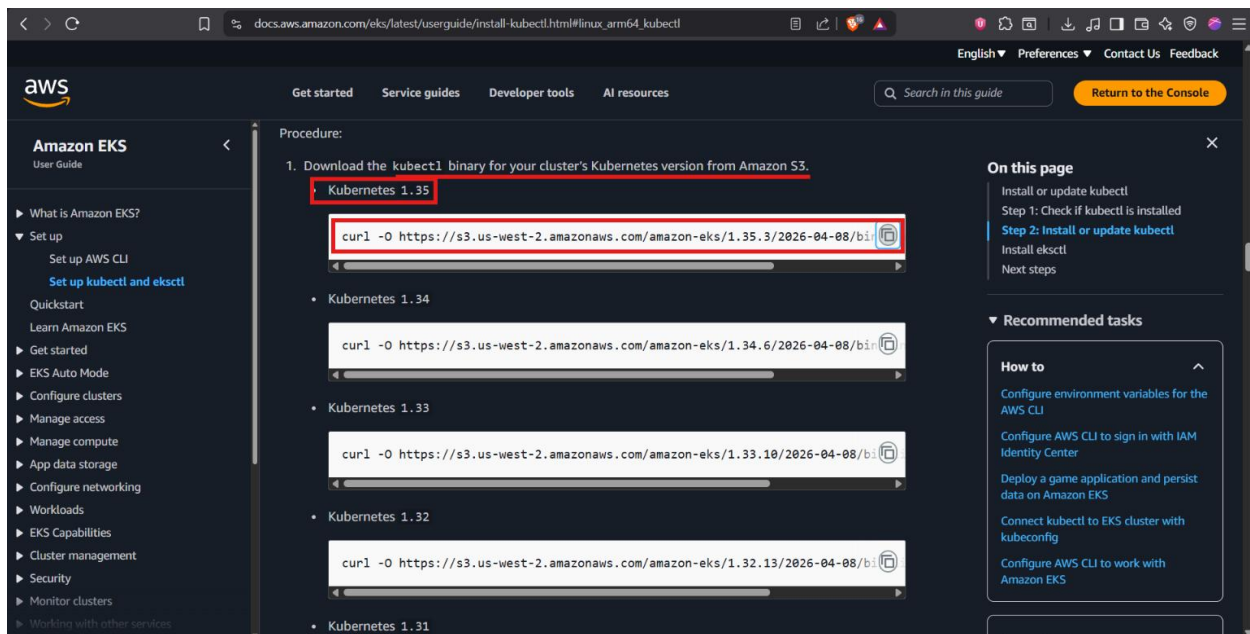
- Switch to root access if required for the setup.

```
sudo -i
```



Step 3 – Verify AWS CLI and configure authentication

- Check that AWS CLI is installed by running the version command.



- Download the kubectl binary.

<https://docs.aws.amazon.com/eks/latest/userguide/install-kubectl.html>

- Give it executable permission.

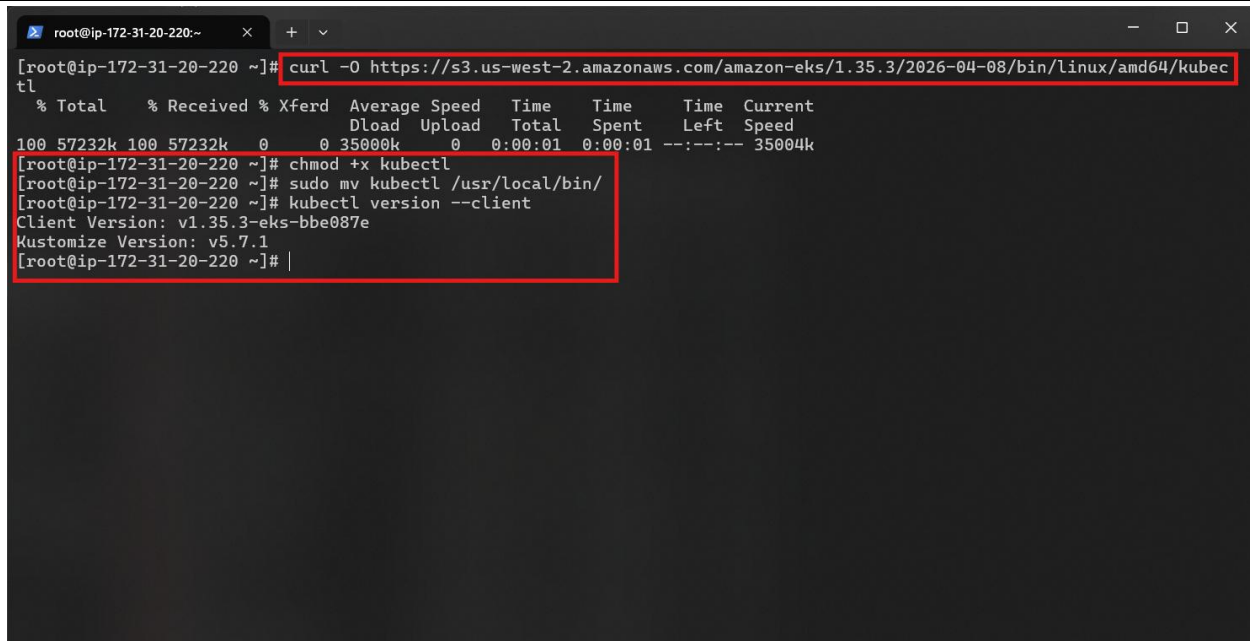
```
chmod +x kubectl
```

- Move it to /usr/local/bin.

```
sudo mv kubectl /usr/local/bin/
```

- Confirm the installation by checking the **kubectl** version.

```
kubectl version --client
```



Step 5 – Generate kubeconfig for the EKS cluster

- Run the AWS CLI command to update kubeconfig for the cluster.

```
aws eks update-kubeconfig --name <YOUR-CLUSTER-NAME> --region <YOUR-REGION>
```

- Use the cluster name and the ap-south-1 Region.
- Confirm that the kubeconfig file was created successfully.

```
[root@ip-172-31-20-220 ~]# aws eks update-kubeconfig --name MyCluster --region us-east-1
Added new context arn:aws:eks:us-east-1:463646775279:cluster/MyCluster to /root/.kube/config
[root@ip-172-31-20-220 ~]#
```

- Understand that this file allows **kubectl** to connect to the EKS API server.

Step 6 – Verify connectivity with kubectl

- Run the following commands

```
kubectl cluster-info
```

```
kubectl get ns
```

```
kubectl get pods -n kube-system
```

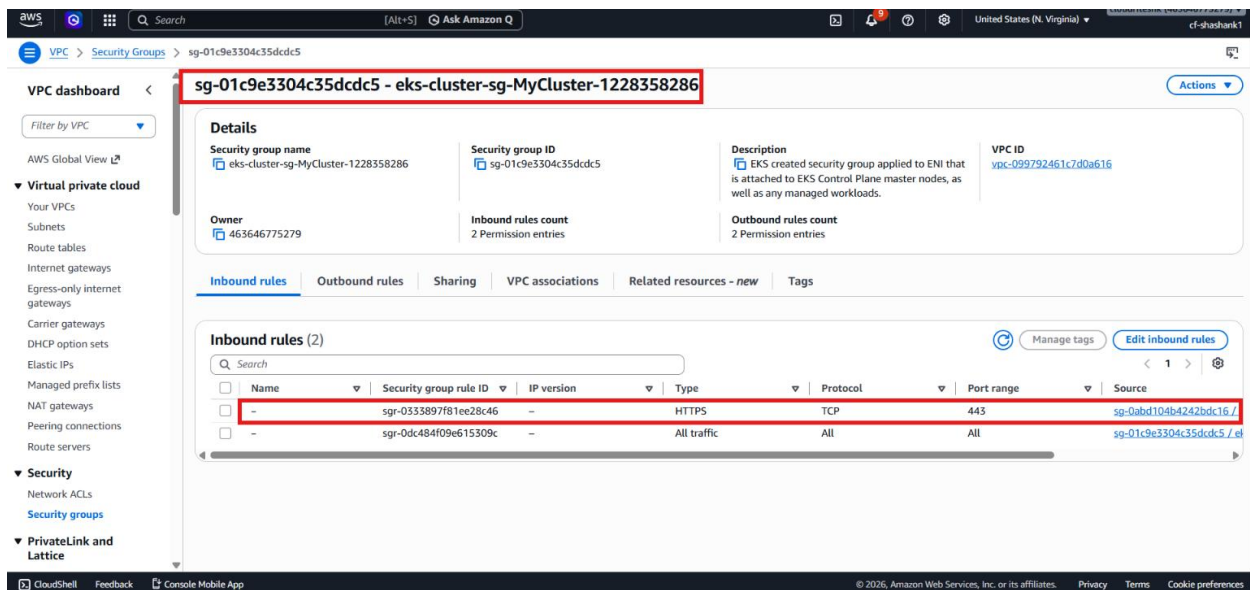
```
kubectl get nodes
```

- Confirm that **kubectl** is talking to the EKS cluster correctly.

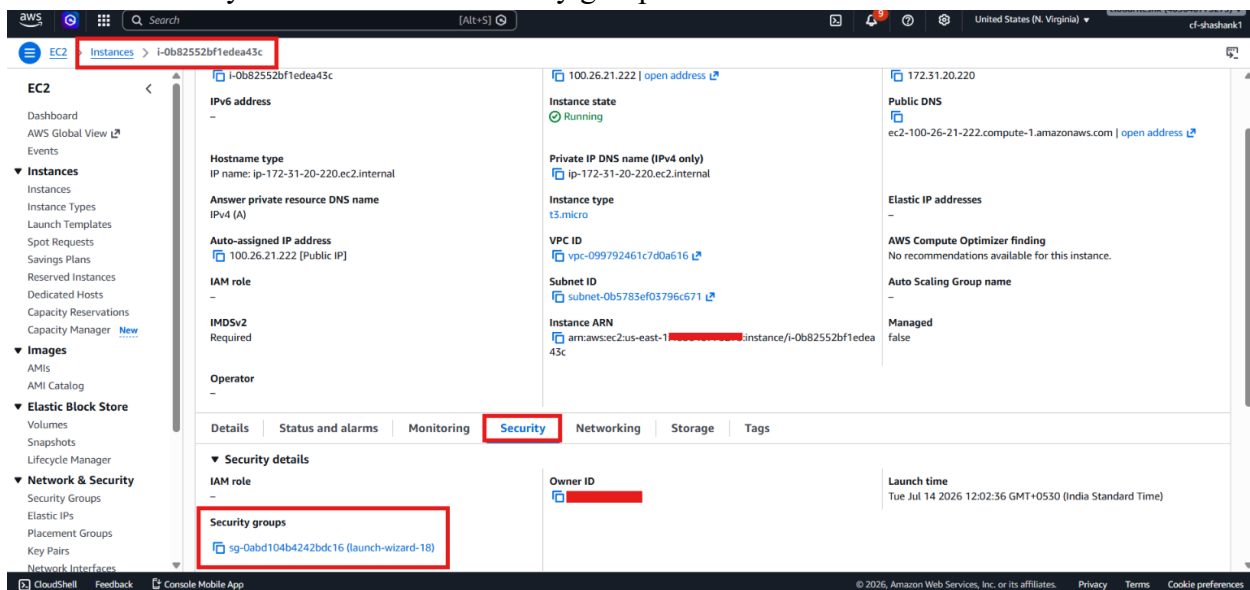
```
root@ip-172-31-20-220:~  
[root@ip-172-31-20-220 ~]# kubectl cluster-info  
Kubernetes control plane is running at https://7B67EA5298FFE373491E83145207155E.gr7.us-east-1.eks.amazonaws.com  
  
To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.  
[root@ip-172-31-20-220 ~]# kubectl get ns  
NAME          STATUS   AGE  
default       Active   77m  
kube-node-lease Active   77m  
kube-public   Active   77m  
kube-system   Active   77m  
[root@ip-172-31-20-220 ~]# kubectl get pods  
No resources found in default namespace.  
[root@ip-172-31-20-220 ~]# kubectl get pods -n kube-system  
NAME                                READY   STATUS    RESTARTS   AGE  
metrics-server-595bd7d7fc-bbkxh    1/1     Running   0           67m  
metrics-server-595bd7d7fc-p822v    1/1     Running   0           67m  
[root@ip-172-31-20-220 ~]# kubectl get nodes  
NAME                                STATUS   ROLES    AGE   VERSION  
i-05380f1739c6d9620 Ready    <none>   68m   v1.36.0-eks-75e0327  
[root@ip-172-31-20-220 ~]#
```

Step 7 – Troubleshoot if the connection fails

- Open the EKS cluster networking section.
- Check the cluster security group.



- Make sure the inbound rule allows access from 0.0.0.0/0 or the correct source.
- Make sure your EC2 instance is in the same VPC.
- Add an inbound rule on the cluster security group
 - Type: HTTPS
 - Protocol: TCP
 - Port: 443
 - Source: <your EC2 instance security group>



- Retry the **kubectl** command after fixing the security group.

Verification

- Verified that the management **EC2** instance was launched in the same **VPC** as the EKS cluster.

- Verified that **AWS CLI** was installed and configured.
- Verified that **kubectl** was downloaded and installed successfully.
- Verified that the kubeconfig file was created with `aws eks update-kubeconfig`.
- Verified that `kubectl cluster-info` returned the EKS control plane details.
- Verified that `kubectl get ns` and `kubectl get pods` returned cluster information.
- Verified that the cluster security group can be checked if connectivity fails.